

```
24 | net.stop()
25
26 if __name__ == '__main__':
27     setLogLevel('info')
28     customTopo()
29
```

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

```
○ amir@Legion-5-Pro-16ITH6:~/Desktop$ sudo python3 project3.py
*** Configuring hosts
h1 h2 h3
*** Starting controller
c0
*** Starting 3 switches
s1 s2 s3 ...
*** Starting CLI:
mininet> 
```

```
○ amir@Legion-5-Pro-16ITH6:~/Desktop$ sudo python3 project3.py
*** Configuring hosts
h1 h2 h3
*** Starting controller
c0
*** Starting 3 switches
s1 s2 s3 ...
Ping from h1 to h2:
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=0.527 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=0.154 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=0.110 ms

--- 10.0.0.2 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2051ms
rtt min/avg/max/mdev = 0.110/0.263/0.527/0.187 ms

Ping from h1 to h3:
PING 10.0.0.3 (10.0.0.3) 56(84) bytes of data.
64 bytes from 10.0.0.3: icmp_seq=1 ttl=64 time=0.856 ms
64 bytes from 10.0.0.3: icmp_seq=2 ttl=64 time=0.117 ms
64 bytes from 10.0.0.3: icmp_seq=3 ttl=64 time=0.113 ms

--- 10.0.0.3 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2044ms
rtt min/avg/max/mdev = 0.113/0.362/0.856/0.349 ms

Ping from h2 to h3:
PING 10.0.0.3 (10.0.0.3) 56(84) bytes of data.
64 bytes from 10.0.0.3: icmp_seq=1 ttl=64 time=0.115 ms
64 bytes from 10.0.0.3: icmp_seq=2 ttl=64 time=0.146 ms
64 bytes from 10.0.0.3: icmp_seq=3 ttl=64 time=0.062 ms

--- 10.0.0.3 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2043ms
rtt min/avg/max/mdev = 0.062/0.107/0.146/0.034 ms

*** Starting CLI:
mininet> █
```

Testing connectivity with ping:

PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.

64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=0.497 ms

64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=0.051 ms

64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=0.067 ms

--- 10.0.0.2 ping statistics ---

3 packets transmitted, 3 received, 0% packet loss, time 2038ms

rtt min/avg/max/mdev = 0.051/0.205/0.497/0.206 ms

PING 10.0.0.3 (10.0.0.3) 56(84) bytes of data.

64 bytes from 10.0.0.3: icmp_seq=1 ttl=64 time=0.389 ms

64 bytes from 10.0.0.3: icmp_seq=2 ttl=64 time=0.048 ms

64 bytes from 10.0.0.3: icmp_seq=3 ttl=64 time=0.042 ms

--- 10.0.0.3 ping statistics ---

3 packets transmitted, 3 received, 0% packet loss, time 2044ms

rtt min/avg/max/mdev = 0.042/0.159/0.389/0.162 ms

PING 10.0.0.3 (10.0.0.3) 56(84) bytes of data.

64 bytes from 10.0.0.3: icmp_seq=1 ttl=64 time=0.086 ms

64 bytes from 10.0.0.3: icmp_seq=2 ttl=64 time=0.105 ms

64 bytes from 10.0.0.3: icmp_seq=3 ttl=64 time=0.044 ms

--- 10.0.0.3 ping statistics ---

3 packets transmitted, 3 received, 0% packet loss, time 2046ms

rtt min/avg/max/mdev = 0.044/0.078/0.105/0.025 ms

Testing bandwidth with iperf:

Between h1 and h2:

*** Iperf: testing UDP bandwidth between h1 and h2

*** Results: ['10M', '10.5 Mbits/sec', '10.5 Mbits/sec']

Between h1 and h3:

*** Iperf: testing UDP bandwidth between h1 and h3

*** Results: ['10M', '10.5 Mbits/sec', '10.5 Mbits/sec']

Between h2 and h3:

*** Iperf: testing UDP bandwidth between h2 and h3

*** Results: ['10M', '10.5 Mbits/sec', '10.5 Mbits/sec']

*** Starting CLI:

mininet> █